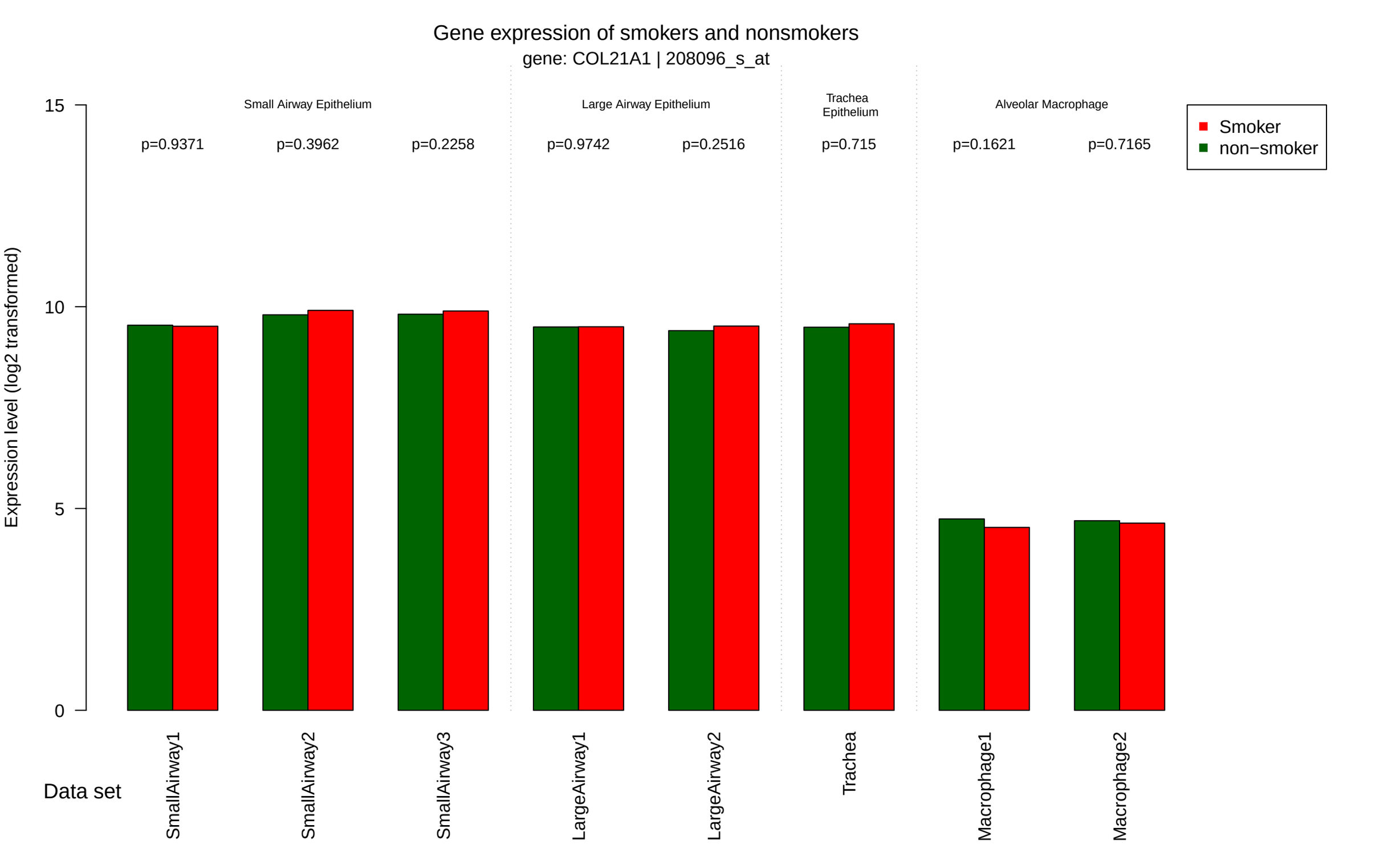
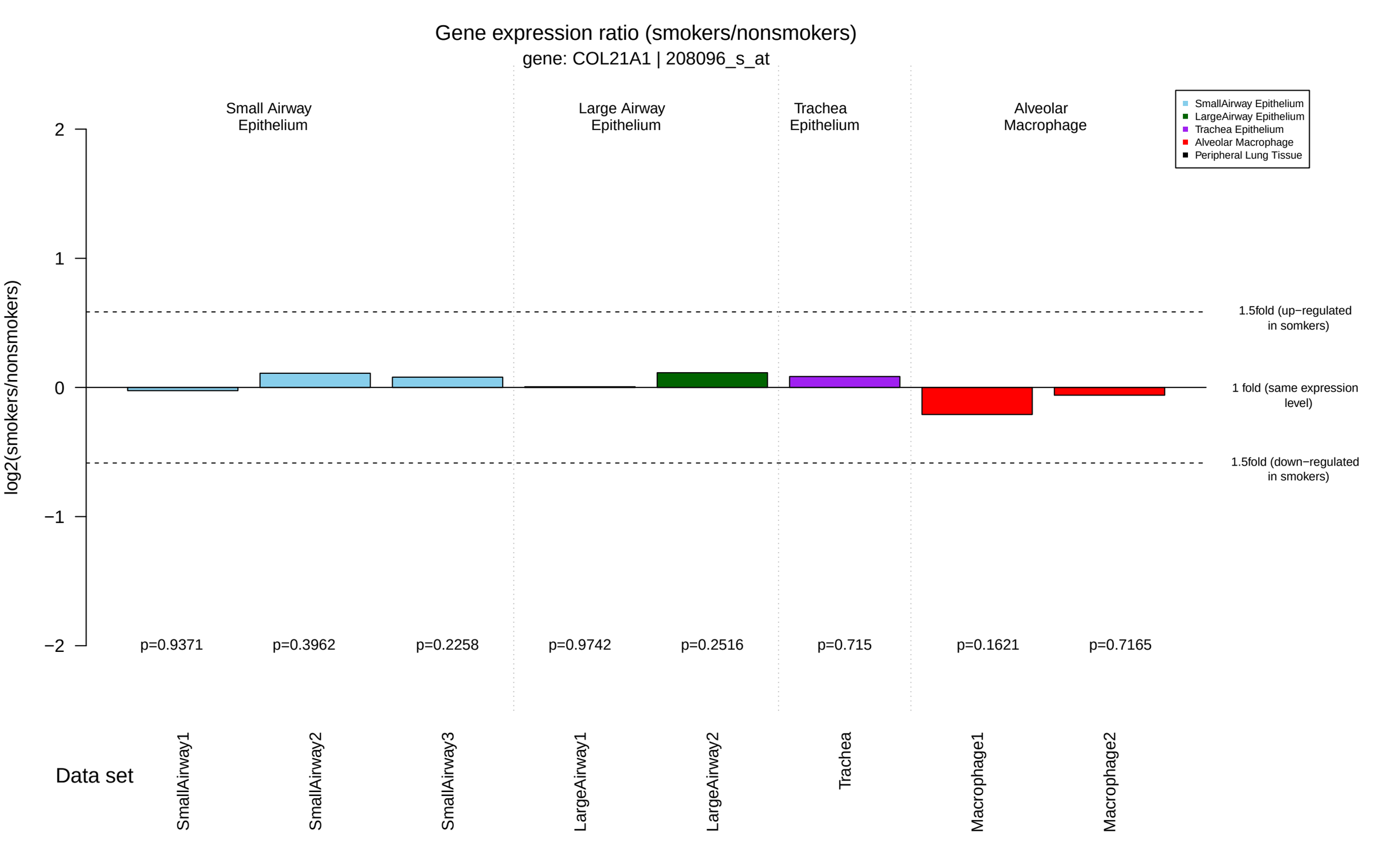
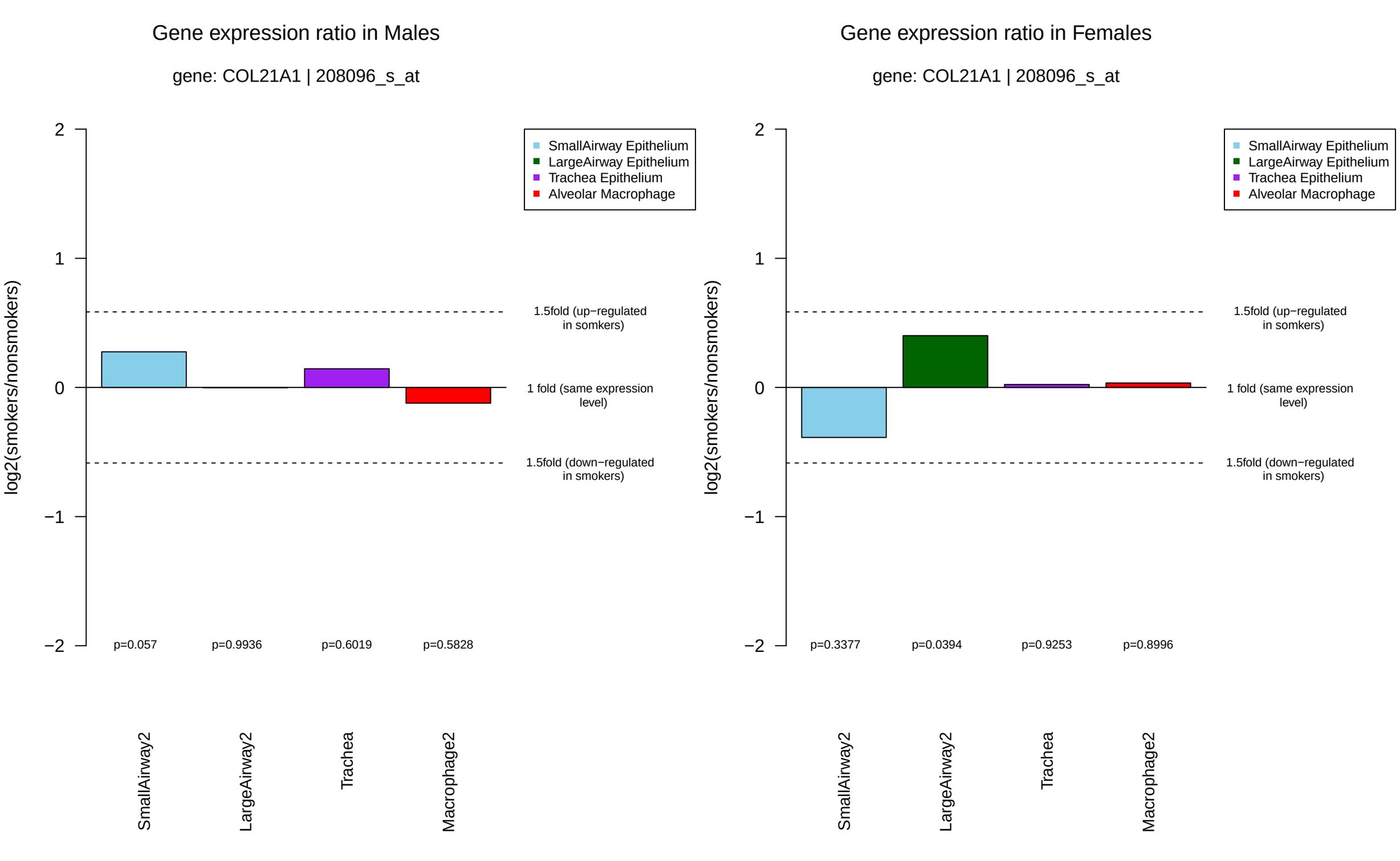
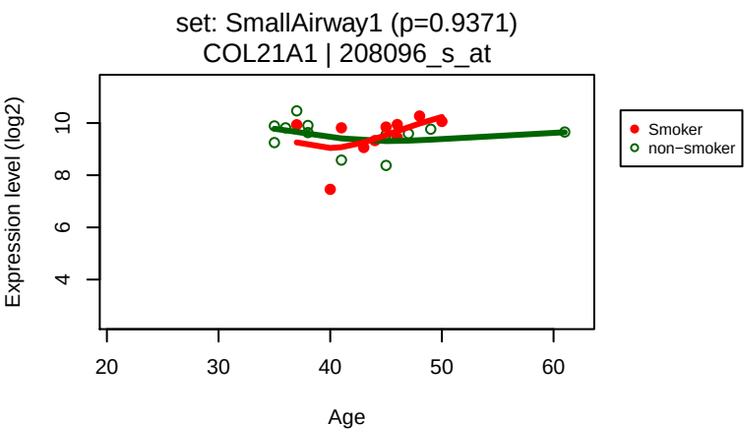
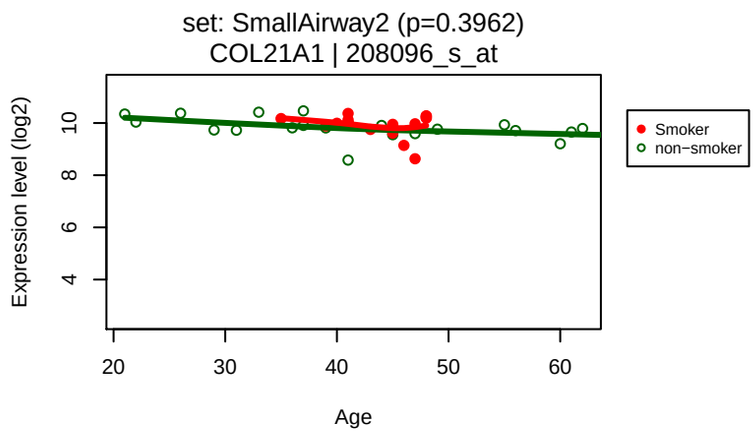
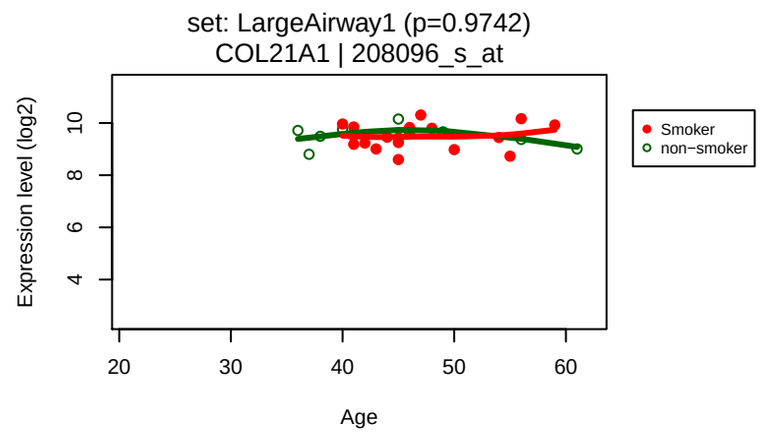

Gene expression of smokers and nonsmokers
gene: COL21A1 | 208096_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9371
p=0.3962
p=0.2258
p=0.9742
p=0.2516
p=0.715
p=0.1621
p=0.7165
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

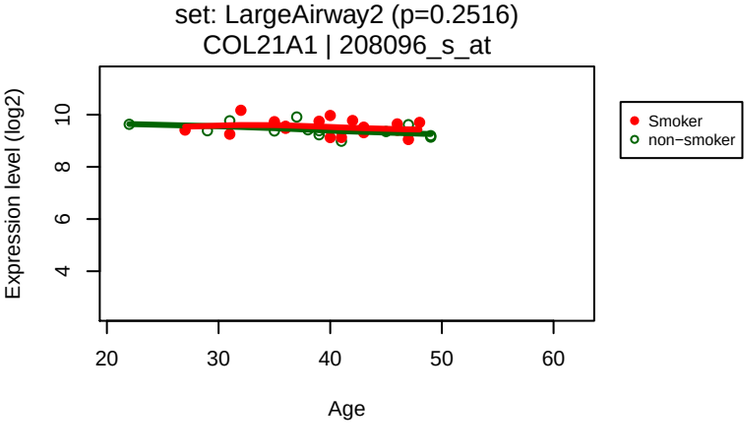

Gene expression ratio (smokers/nonsmokers)
gene: COL21A1 | 208096_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9371
p=0.3962
p=0.2258
p=0.9742
p=0.2516
p=0.715
p=0.1621
p=0.7165
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

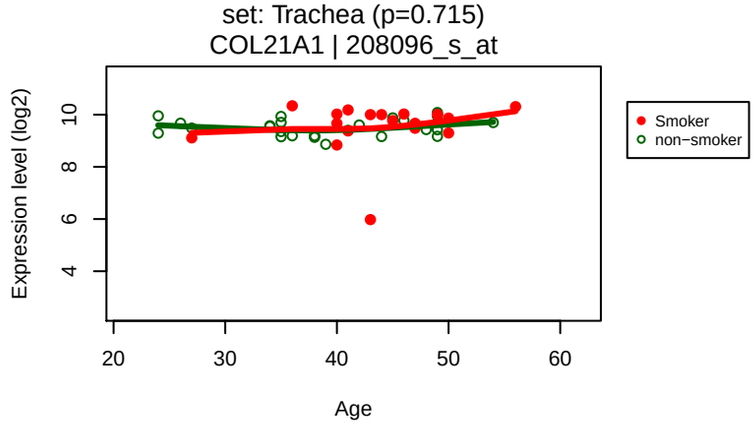

Gene expression ratio in Males
gene: COL21A1 | 208096_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.057
p=0.9936
p=0.6019
p=0.5828
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL21A1 | 208096_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3377
p=0.0394
p=0.9253
p=0.8996
SmallAirway2
LargeAirway2
Trachea
Macrophage2

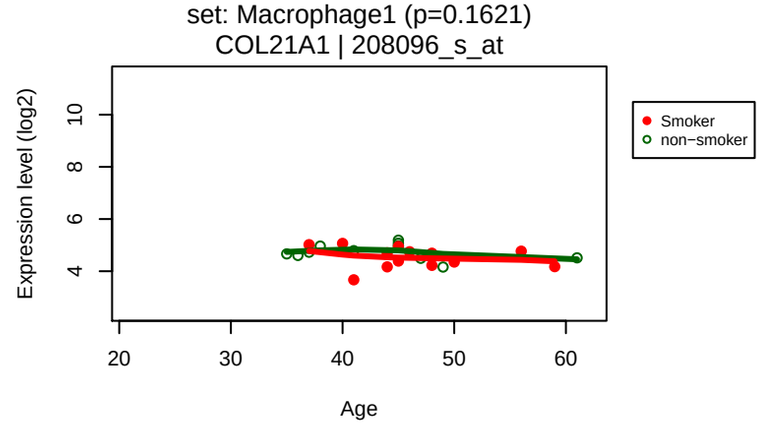

set: SmallAirway1 (p=0.9371)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker

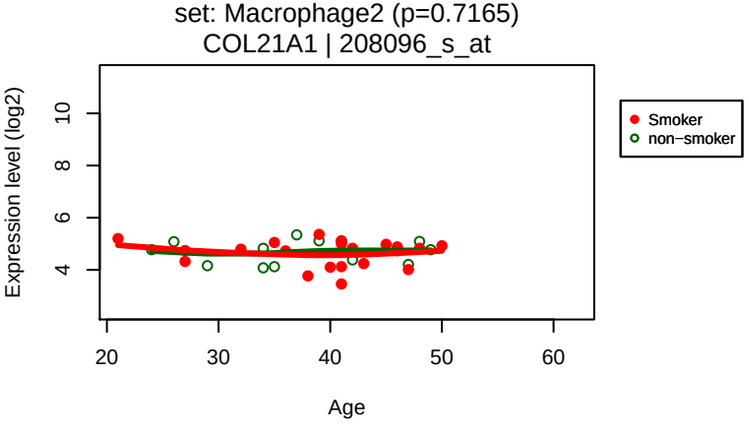

set: SmallAirway2 (p=0.3962)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker

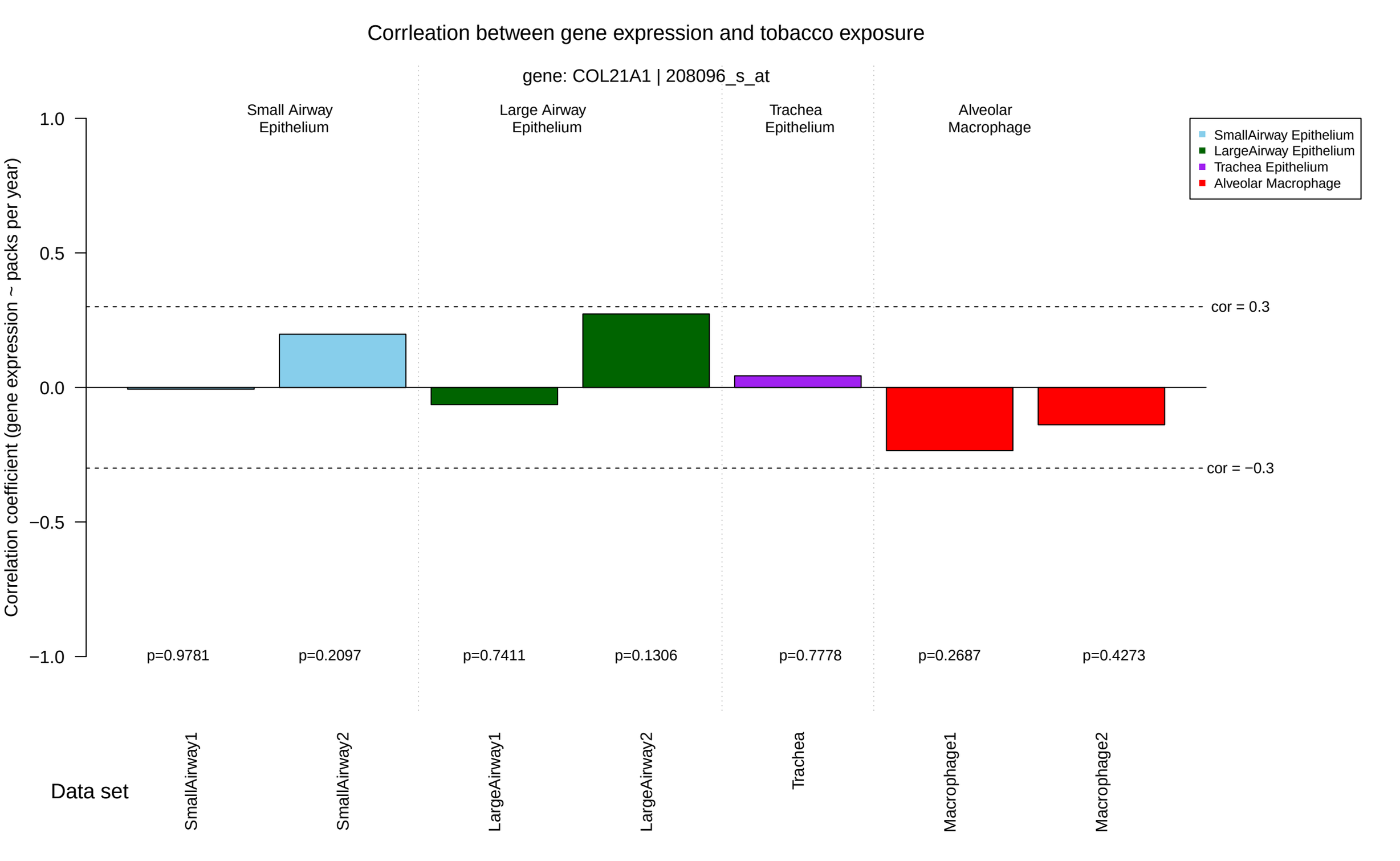

set: LargeAirway1 (p=0.9742)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2516)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.715)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1621)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7165)
COL21A1 | 208096_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL21A1 | 208096_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9781
p=0.2097
p=0.7411
p=0.1306
p=0.7778
p=0.2687
p=0.4273
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2